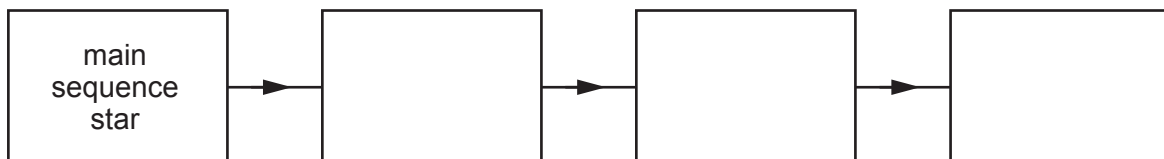


3. Rigel is a high-mass blue star, 870 light-years away from Earth.
Rigel is currently on the main sequence.

- (a) Use an equation from page 2 to calculate the distance of Rigel from Earth in metres. [3]
Speed of light, $c = 3 \times 10^8$ m/s
1 year = 31 536 000 s

distance = m

- (b) (i) Complete the boxes to show the remaining stages in Rigel's life cycle. [3]



- (ii) Explain why large mass stars are important in the creation of new solar systems. [2]

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- (iii) Describe how a solar system forms. [2]

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